

Course

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1.1.1. Calculate Momentum

01:19

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:
$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

Submit

1 mass=float(input())
2 velocity=float(input())
3 p=mass*velocity
4 print(f"{p:.2f}kgm/s")

Average time 0.009 s 8.60 ms
Maximum time 0.011 s 11.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 11 ms Debug

Expected output
5.0
10.0
50.00kgm/s

Actual output
5.0
10.0
50.00kgm/s

Test case 2 8 ms

Terminal Test cases

Breaking news
VCK, IJML give...

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1.1.2. Conditional Calculation Based on the Number of Digits

03:11

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 # n = int(input())
2 # if n < 1 or n >= 1000:
3 #     print("Invalid")
4 # elif n < 10:
5 #     print(n**2)
6 # elif n < 100:
7 #     print(f"{n**0.5:.2f}")
8 # else:
9 #     print(f"{n**(1/3):.2f}")
10 n=int(input())
11 if n>0 and n<=9:
```

Average time
0.005 s
5.00 ms

Maximum time
0.008 s
8.00 ms

4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 8 ms

Expected output
9
81

Actual output
9
81

Test case 2 4 ms

Test case 3 6 ms

Terminal Test cases

Hot days ahead
34°C

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`.
- The second line contains the second date in the format `YYYY-MM-DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```
1 # from datetime import date
2
3 # #write your code here...
4 from datetime import date
5 date1 = date.fromisoformat(input())
6 date2 = date.fromisoformat(input())
7 print(abs(date1-date2).days)
```

Average time

0.005 s

5.00 ms

Maximum time

0.006 s

6.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

5 ms

Debug

Expected output

Actual output

2014-07-02

2014-07-02

2014-11-02

2014-11-02

123

123

Test case 2

4 ms

Terminal

Test cases

Hot days ahead

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1.1.4. Reverse a Number00:11

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 #write your code here...
2 num=int(input())
3 num2=0
4 while num!=0:
5     digit= num% 10
6     num2=(num2*10)+digit
7     num= num//10
8 print(num2)
```

Average time0.003 s3.00 ms

Maximum time0.004 s4.00 ms

2 out of 2 shown test case(s) passed3 out of 3 hidden test case(s) passed

Test case 14 ms

Expected output53677635

Actual output53677635

Test case 22 ms

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1.1.5. Multiplication Table

00:05

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multipl...

1#write your code here...
2
3
4a=int(input())
5for i in range (1,11):
6print(f"{a} x {i} = {i*a}")

Average time0.003 s2.75 msMaximum time0.004 s4.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 13 msDebug

Expected output
8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40

Actual output
8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40

TerminalTest cases

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

passorFa...

1

write your code here

2

n = int(input())

3

marks = list(map(int, input().split()))

4

5

if min(marks) < 40:

6

print("Fail")

7

else:

8

percentage = sum(marks) / n

9

print(f"Aggregate Percentage: {percentage:.2f}")

10

11

if percentage > 75:

Average time

0.007 s

7.30 ms

Maximum time

0.010 s

10.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1

10 ms

Debug

Expected output

4

55 65 78 89

Aggregate Percentage: 71.75

Grade: First Division

Actual output

4

55 65 78 89

Aggregate Percentage: 71.75

Grade: First Division

Test case 2

8 ms

Terminal

Test cases

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 # def fibonacci(n):
2
3 # write the code..
4
5 # return fibonacci( )
6
7
8 # n = int(input())
9 # for i in range(1, n + 1):
10 # print(fibonacci(i), end=" ")
11 # def fibonacci(n):
```

Average time
0.009 s
8.60 ms

Maximum time
0.021 s
21.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 4 ms

Expected output
0 1 1 2 3

Actual output
0 1 1 2 3

Test case 2 4 ms

Terminal Test cases

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 #Type Content here...
2 n = int(input())
3 for i in range(1, n+1):
4     print("*" * i)
```

Average time0.003 s3.17 msMaximum time0.005 s5.00 ms

2 out of 2 shown test case(s) passed4 out of 4 hidden test case(s) passed

Test case 13 msDebug

Expected outputActual output

```
5
*
* *
* * *
* * * *
* * * * *
```

TerminalTest cases

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